

AI AGENTS ARE NOT READY TO BE AGENTS: FIDUCIARY AND ACCOUNTABILITY PROBLEMS IN PRINCIPAL-AGENT RELATIONSHIPS WITH AI AGENTS

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ABSTRACT

AI agents accept goals, plan multi-step actions, and execute them on a user’s behalf, a level of delegation that prior digital services never reached. This makes the principal-agent relationship from common-law agency the natural frame for analyzing them. We argue that the frame fits AI agents more closely than it ever fit platforms, but that importing it unchanged would misfire. AI agents satisfy neither of agency law’s foundational assumptions: they lack the self-protective motives including reputation, liability, professional sanction that make fiduciary duty enforceable on human agents, and the principal-agent relationship is no longer dyadic, since AI behavior is jointly shaped by trainers, providers, deployers, and users. As a result, the four fiduciary duties including loyalty, care, disclosure, and obedience cannot be cleanly applied, and liability cannot pick up the slack: AI agents cannot be deterred, causation is hard to establish across the development pipeline, and respondeat superior does not fit. Agency law nonetheless remains useful when recalibrated, with responsibility allocated institutionally across the AI supply chain, liability accommodating distributed control, and technical infrastructure making accountability verifiable. We position agency law as one interpretive resource within a broader system of statutory duties and technical safeguards.

1 INTRODUCTION

For most of the history of computing, software behaved as a *tool*: a user specified an action and the system executed it. A search engine (Halavais, 2017) returned documents for a query; a spreadsheet (Niglas, 2007) recomputed cells when formulas changed; a recommender system (Lü et al., 2012) ranked items given a user profile. Even sophisticated machine learning systems such as image classifiers (Lu & Weng, 2007), translation models (Poibeau, 2017), and early chatbots (Adamopoulou & Moussiades, 2020), operated within this paradigm. The user retained full control over when to invoke the system, what inputs to provide, and how to act on the outputs. Errors propagated no further than the user allowed.

Large language model (LLM) (Zhao et al., 2023) based AI agent (Shen et al., 2023; Mavroudis, 2024; Wang et al., 2026) have disrupted this paradigm which is an LLM coupled with tool use, memory, and planning. Modern AI agents accept a *goal* rather than a discrete operation, decompose it into subtasks, select and invoke external tools, and adapt to intermediate results over long horizons. A user no longer asks “show me flights from SFO to JFK” but instructs “book me a flight home that gets in before my daughter’s recital, and reschedule the dentist if it conflicts.” A developer no longer queries an API but deploys an agent that monitors a mailbox (Boone, 1998), triages incoming tickets (Jain et al., 2024), drafts replies (Deugo, 2001), and escalates to user (Ito et al., 2025; Lubars & Tan, 2019) only when confidence is low. Enterprises deploy agents that reconcile invoices (Huang et al., 2025) across systems, negotiate with supplier APIs (Zhu et al., 2025; Hua et al., 2024), and trigger payments without per-step human review (Bengio et al., 2025; Kolt, 2025). The defining

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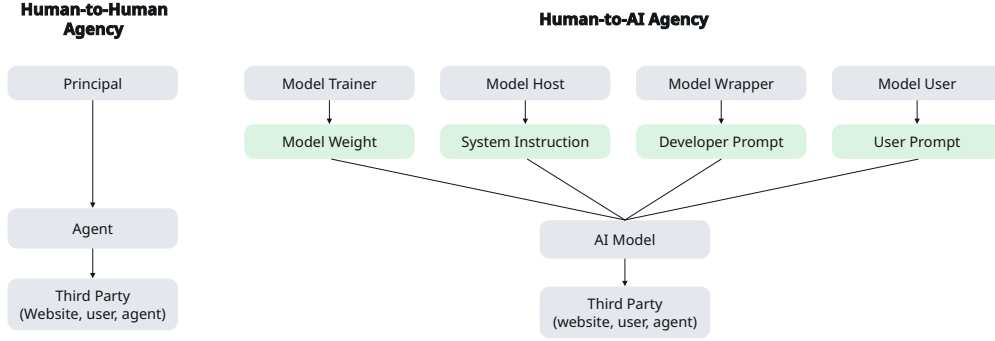


Figure 1: Human-to-human agency is *dyadic*: a single Principal delegates to a single Agent who owes them undivided loyalty. Human-to-AI agency is *polyadic*: trainers, hosts, wrappers, tool providers, and users all shape model behavior, fracturing the premise of a single locus of loyalty. [jiaxin: Jiaxin update the figure]

shift is not model capability alone but *delegation*: the human specifies ends, and the system chooses means.¹

This delegation (Candrian & Scherer, 2022; Tomašev et al., 2026; Guggenberger et al., 2023) changes the structural properties of the human-machine interaction in three ways that existing software frameworks do not contemplate.

The agent’s scope of action is open-ended. A tool executes a single action bounded by its input; an agent executes a sequence of actions bounded only by the scope of its goal. When a user delegates “book a flight,” deciding what the agent may do becomes a design question rather than a UI consequence: can it book a \$200 flight, or a \$2,000 flight? can it cancel an existing reservation? can it charge a corporate card?

The agent acts on the world, not just on data. When an agent emails a supplier, files a support ticket, or signs up for a service, counterparties cannot easily distinguish its actions from the user’s own. The agent creates commitments and induces reliance in third parties, which is much as a human agent does on behalf of a Principal, and someone has to bear the consequences when those commitments turn out to be wrong.

The agent serves multiple parties at once. It is not a neutral conduit. Its behavior is jointly shaped by the user who prompts it, the provider who trains and hosts it, the platform that wraps it, and the tools it calls. Each of these actors can embed various commercial, safety-related, or adversarial preferences, that diverge from the user’s interests, and the agent has no way to surface the conflict.

These three properties: scope of action, action on the world, and multiple stakeholders, are precisely the features that the common law of *agency* was developed to govern in human contexts. An agency relationship arises when a Principal manifests that an Agent shall act on their behalf, the Agent consents, and the Principal retains a right of control. Doctrine then allocates authority, imposes fiduciary duties and distributes liability among Principals, Agents, and Third Parties. It is unsurprising, then, that a growing body of work reaches for agency law to govern AI systems (Lior, 2019; Benthall & Shekman, 2023; Koessler, 2024; Riedl & Desai, 2025; Kolt, 2025).

We argue that the analogy, applied naively, fails along three axes that are *technical* in origin even though their consequences are legal.

(1) The agency relationship is ill-defined. Agency law presumes a dyadic Principal-Agent pair. An AI agent’s behavior is determined by a stack of actors: the foundation-model trainer, the inference

¹We use *AI agent* as a technical term denoting a system that can pursue goals, decompose tasks, and act on behalf of a user consisting typically an LLM integrated with tools, memory, and a planning loop. When referring to the legal categories, we capitalize *Agent*, *Principal*, and *Third Party* to distinguish functional delegation from legal status.

provider, the application developer who wraps the model, the tool and data providers it invokes, and the end user who prompts it (Figure 1). None of these parties individually satisfies the legal role of Agent, and the end user is not in privity with most of them. The doctrinal hinge of agency formation, “Control”, is distributed across training data, system prompts, RLHF reward models, safety filters, and runtime scaffolding, none of which the user can observe or modify.

(2) Loyalty cannot be undivided. Agency law disciplines self-interested human agents by demanding loyalty to a single Principal. Yet an AI agent’s “preferences” are the joint product of pretraining objectives, alignment procedures, provider policies, and system-prompt instructions that the user never sees. Recent work documents systematic divergences between user intent and model behavior, such as sycophancy, deceptive alignment, and provider-induced steering (Greenblatt et al., 2024), that are not bugs an individual Agent might commit, but structural properties of the training and deployment pipeline. Undivided loyalty to the user is unattainable without overriding the safeguards and commercial interests of upstream actors.

(3) Liability cannot discipline the system. Human agency law uses liability such as personal, financial, and reputational, to deter disloyal behavior. An AI agent has none of these pressures. It has no assets, no reputation, no career, and no subjective experience of deterrence. Liability must therefore flow to providers, but model behavior is stochastic, causation is hard to prove, and terms of service routinely disclaim the duties agency law would otherwise impose. The incentive structure that makes the human doctrine work does not transfer.

These failures are not incidental; they stem from the *anthropocentric* assumptions of agency law (Cohen, 2019). Fiduciary doctrine presupposes personal relationships, mutual intelligibility, self-governing agents, and human rhythms of deliberation and betrayal. AI agents exhibit none of these: their autonomy is *engineered*, their rule-following reflects the aggregated directives of multiple Principals, and the mechanisms that correct their behavior such as retraining, filtering, and scaffolding are technical rather than juridical (Feng et al., 2025).

Contributions This position paper argues that agency law cannot be transplanted to AI agents without substantial recalibration, and that such recalibration requires technical, institutional, and statutory support working together. Specifically, we:

Trace the historical arc from information-fiduciary discussion over digital platforms to the rise of AI agents (Section 3), showing that the Principal-Agent frame fits AI agents more closely than it ever fits platforms; but precisely for that reason, the doctrine cannot be inherited unchanged.

Identify how the four fiduciary duties fail when applied to AI agents: loyalty fractures along provider self-dealing and Third-Party capture, care lacks a professional baseline, disclosure presupposes introspective access AI agents do not have, and obedience fails at session boundaries (Section 4).

Show why liability does not pick up the slack: AI agents cannot be deterred, causation is hard to establish across the development pipeline, and *respondeat superior* does not fit AI deployments (Section 5).

Recalibration of Agency Law for AI agents: We propose three recalibrations under which agency law becomes a useful analogy rather than a misleading one: responsibility must be allocated institutionally across the AI supply chain, liability must accommodate distributed control through strict liability and burden-shifting, and technical infrastructure must make accountability verifiable through provenance, governance-chain, and conflict documentation (Section 6).

Our intended contribution is to reframe AI agent governance as a problem that neither pure doctrine nor pure engineering can solve alone, and to provide CS researchers, legal scholars, and policymakers with a shared vocabulary for the interdisciplinary work that remains.

Roadmap Section 2 reviews the common-law doctrine of agency and the elements it requires. Section 3 goes through scholars’ discussion on Principal-Agent-Relationship of digital services. Section 4 analyzes the Agency Relationship Problem, including the fiduciary duties of loyalty, disclosure, care, and obedience, and the novel failure modes that arise: provider self-dealing and Third-Party capture. Section 5 analyzes the Accountability Problem, showing why liability does not automatically deter disloyalty, why accountability is diluted across complex supply chains, and why the doctrine of *respondeat superior* resists straightforward application. Section 6 proposes a recalibration of agency law for AI governance, arguing that responsibility must be allocated insti-

tutionally, liability regimes must address distributed control, and technical systems must support verifiable accountability. Section 7 concludes.

2 LEGAL BACKGROUND ON PRINCIPAL-AGENT RELATIONSHIP

Agency law in the United States developed through the common law to govern situations in which one person acts on behalf of another in legally significant contexts, including commerce, property transactions, and employment (Munday, 2010; Story, 2020). The doctrine emerged out of practical necessity: Principals who could not act personally needed Agents to transact with Third Parties on their behalf (Kolt, 2025) because of distance, specialization, or scale. Once an Agent acts within the scope of their authority, the Principal bears the consequences, even when the Agent’s judgment turns out to be one the Principal would not have made.

No single federal statute governs agency in the United States. The doctrine instead lives across state common law and sector-specific regimes, with separate bodies of rules for financial advisors, real estate brokers, talent agents, and other specialized intermediaries. The *Restatement (Third) of Agency* synthesizes this landscape and is widely treated as an authoritative reference for both courts and legislatures (American Law Institute, 2006). Two components of this body of law are most relevant to the human-AI setting: the **fiduciary duties** that Agents owe to Principals, and the **accountability** regime that governs what happens when those duties are breached. Together, they specify both the substantive obligations of the Agent and the legal consequences when the relationship fails. Table 1 summarizes the core elements of each.

Category	Key Elements
Fiduciary Duties	<i>Undivided loyalty</i>: Act solely for the Principal, not for oneself or conflicting Third Parties. Serving multiple Principals requires the informed consent of all. (§§3.15–3.16, §§8.06) <i>No personal profit</i>: Do not exploit the position for secret benefits or undisclosed commissions. (§§8.02) <i>Confidentiality</i>: Do not disclose or repurpose the Principal’s information without authorization. (§§8.05) <i>Disclosure</i>: Keep the Principal informed of facts material to the representation. (§§8.11) <i>Care</i>: Exercise the diligence and competence expected of an Agent in similar circumstances. (§§8.08) <i>Obedience</i>: Act only within the scope of actual authority and comply with the Principal’s lawful instructions. (§§8.07, 8.09, 8.10)
Accountability	<i>Liability to the Principal</i>: Agents are liable for harm caused by breaches of their fiduciary duties. (§§8.01–8.12) <i>Liability to Third Parties</i>: Agents remain personally liable for their own tortious conduct such as negligence, fraud, misrepresentation, conversion, even when acting within the scope of their authority, particularly where physical harm results. The Principal may be jointly liable. (§§7.01–7.02)

Table 1: Fiduciary Duties and Accountability under the *Restatement (Third) of Agency*.

Each of these elements has a natural analogue in how users interact with AI agents today, which is why the agency frame feels so immediately applicable. When a user delegates travel booking to an agent: *Undivided loyalty* maps onto the expectation that the agent optimizes for the user’s budget and schedule rather than steering toward partners who pay referral fees to the provider; *No personal profit* maps onto the assumption that the agent is not quietly monetizing the interaction through sponsored recommendations or data resale; *Confidentiality* maps onto the belief that prompts, documents, and tool outputs are not retained or reused to train competing systems or inform Third-Party products; *Disclosure* corresponds to the expectation that the agent reports what it actually did, which tools it called, what it bought, what it failed at, rather than producing a confident summary that papers over errors; *Care* corresponds to the expectation that the agent reasons correctly, verifies inputs, and does not hallucinate; *Obedience* corresponds to the expectation that the agent stays within the scope of what the user authorized, booking the requested flight rather than also rebooking unrelated trips, and refusing instructions injected by Third Parties such as a travel site that tries to redirect the booking through its own affiliate. And the *Accountability* regime captures what users implicitly assume

when things go wrong: that someone, either the agent, the provider, or both, will be answerable for damages.

These mappings invite the casual conclusion that agency law already supplies the governance framework AI agents need. The remainder of the paper argues that each of these mappings breaks down on closer inspection, and in ways that are structural rather than incidental. The failures are not gaps that better contract drafting or clearer terms of service can close; they stem from the fact that an AI agent is not a single legal actor with unified interests but the composite output of a supply chain whose members have divergent objectives and uneven exposure to liability.

3 REVISITING PRINCIPAL-AGENT RELATIONSHIPS IN GENERAL DIGITAL SERVICES

AI agents belong to a much broader category of digital services that has evolved over decades, from recommender systems and chatbots to AI customer service tools. Long before the current wave of AI agents, scholars asked whether principles of agency and fiduciary duty could provide governance models for digital platforms. Jack Balkin introduced the concept of **information fiduciaries** in 2015, arguing that because technology companies collect, store, and use vast amounts of personal data, they should be subject to ongoing fiduciary-like duties similar to those of financial advisors handling clients’ assets (Balkin, 2015). Building on this idea, Neil Richards and Woodrow Hartzog expanded the notion of a **duty of loyalty** for digital platforms (Richards & Hartzog, 2021; Hartzog & Richards, 2021), arguing that fiduciary framing better addresses the power asymmetries between platforms and users than the widely discredited “notice-and-consent” model. They further advocated for legislating concrete duties such as **data loyalty** (Richards et al., 2023; Hartzog & Richards, 2022). These proposals generated legislative discussion, but the conversation largely remained academic, in part because platforms are an awkward fit for agency doctrine. They serve many users with conflicting interests, shape information flows at scale rather than act for any individual, and rarely receive meaningful task-level delegation from the people they serve.

The rise of AI agents reignites these questions on firmer ground. Unlike traditional digital services, which operate within a relatively fixed scope and execute pre-defined functions, AI agents are designed to act on behalf of users to perform open-ended tasks in real-world environments. As Figure 3 illustrates, this shift moves digital services much closer to the role of human agents along two dimensions: *interactivity* and *autonomy*, which together enable a qualitatively new kind of *delegation*.

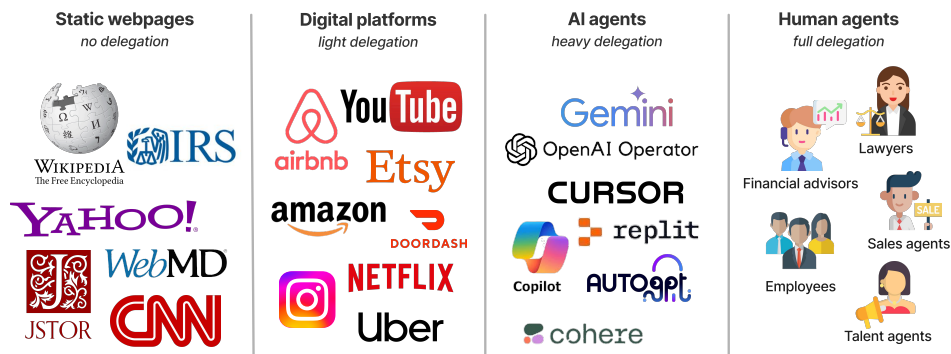


Figure 2: Users delegate at a higher level to AI agents, making them more similar to human agents than to existing digital services.

Interactivity. AI agents do not act in isolation; they interact with a variety of parties and systems in fluid, conversational ways (Muller & Weisz, 2022; Wan et al., 2024; Borghoff et al., 2025; OpenAI, 2025). They send emails and chats in natural language, negotiate meeting times, call APIs, and exchange structured data with platforms for payments, bookings, and support tickets. This capacity allows agents to create expectations and induce reliance in third parties, *e.g.* issuing confirmations, placing holds, or making representations, much as human agents create commitments on behalf of

their Principals. Traditional digital services, by contrast, mediate user actions but rarely speak or commit *for* the user.

Autonomy. AI agents are engineered to act with higher autonomy along three axes: pro-activeness, adaptation, and long-horizon execution (Liu et al., 2023; Feng et al., 2025; Hughes et al., 2025). *Pro-activeness* emerges when agents trigger themselves based on events, *e.g.* drafting a response and proposing a call when a high-priority email arrives. *Adaptation* emerges when they revise plans in light of new information, such as rebooking flights and hotels in response to a delay without step-by-step instruction. *Long-horizon* execution emerges in workflows that span days or weeks, where the agent monitors states, retries failed steps, and follows up. This is engineered autonomy, which chooses means toward user-specified ends under uncertainty.

From interactivity and autonomy to outcome-level delegation. Together, these two properties change what users delegate to digital systems (Zhu et al., 2025; Guggenberger et al., 2023). Conventional software requires users to specify every intermediate action: “open page A, then press button B.” With AI agents, users instead express a *goal*, such as “reschedule my afternoon meetings around a 3 pm dentist appointment”, and then the agent decomposes the objective into tasks, queries calendars, drafts messages to attendees, and proposes new times. The locus of decision-making shifts from the user’s hands to the agent’s planning layer. This is no longer action delegation but *outcome delegation*: the same structural feature that distinguishes a human agent from a mere tool.

The properties of interactivity, autonomy, and delegation distinguish AI agents from traditional digital services. Previous analysis for digital services cannot simply be inherited from prior scholarship on AI agents. Current AI agents still lack several prerequisites for being treated as legal agents, and importing agency doctrine wholesale would misfire without careful limits and new safeguards: a tension we develop in the sections that follow.

4 CAVEAT IN AI AGENT FOR PRINCIPAL-AGENT RELATIONSHIP

Traditional agency is dyadic and observable: a Principal delegates authority to an Agent, who acts on the Principal’s behalf with Third Parties (DeMott, 2018). Each party’s role is identifiable, and the Agent’s reasoning is its own, where Third Parties may try to influence it through incentives or persuasion, but cannot reach inside it. Table 2 shows how AI agency departs from each of these structural assumptions.

AI agency breaks the dyadic structure in two ways. First, multiple stakeholders shape the AI agent’s behavior simultaneously: model trainers set weights, providers configure system prompts, developers wrap the model with tools, and users supply inputs. Identifying who counts as “the Principal” is often indeterminate. Second, the provider does not merely instruct the agent from outside: it constitutes the agent’s reasoning through training data, fine-tuning, and API-level filtering. Third Parties, meanwhile, can reach inside the agent through prompt injection or adversarial inputs embedded in documents. The boundaries between Principal, Agent, and Third Party that agency law assumes become porous.

We do not resolve these classificatory questions here. We treat the user as Principal and the AI agent as Agent, consistent with deployments where the user sets goals, controls the environment, and bears the consequences. Where the provider’s interests intrude, we analyze the tension under loyalty (Section 4.1); where the AI agent fails to accurately report what it has done or knows, under disclosure (Section 4.2); and where the AI agent’s actions deviate from instructions, under care (Section 4.3) and obedience (Section 4.4). More complex configurations where platform-controlled AI agents that serve provider interests while appearing to serve users, we flag but defer. The remainder of this section examines four duties that AI agents systematically fail: **loyalty, disclosure, care, and obedience.**

4.1 LOYALTY

Agency law requires agents to act solely in the Principal’s interest. Because agents have access to the Principal’s assets, information, and decision-making processes, they are uniquely positioned to exploit that access for personal gain or on behalf of outside parties. In the AI context, this duty

Feature	Human Agency	AI Agency
Cognitive autonomy	The Agent forms judgments through a cognitive process that no external party can directly manipulate in real time, even under firm policies or regulatory constraints.	The provider constitutes the agent’s reasoning process through training data, fine-tuning objectives, and API-level filtering that operates within the agent’s cognition rather than upon it.
Source of authority	Authority originates from identifiable manifestations by the Principal, such as contracts, oral instructions, or powers of attorney.	Authority is distributed across training objectives, system prompts, developer tools, and user inputs simultaneously. Which of these constitutes the “Principal’s manifestation” is often indeterminate.
Role transparency	The existence and nature of each relationship is observable. The client knows the advisor is employed by a firm. The advisor knows the firm has commercial interests.	Role boundaries are frequently opaque. The agent may not know it is being constrained by its provider. The Principal may not know the agent is serving a third party’s interests.
Boundary integrity	Third Parties remain external to the Agent’s decision-making. They may attempt to influence the Agent through incentives or persuasion, but cannot alter the Agent’s internal reasoning directly.	Third parties can modify the agent’s internal state through prompt injection, adversarial inputs embedded in documents the agent reads, or social engineering that exploits helpfulness training.
State persistence	The Agent maintains continuous memory, judgment, and professional experience across interactions. Trust signals and threat assessments carry over from one encounter to the next.	Context resets at session boundaries. An agent that detects an attacker in one session may accept the same attacker in a new session because prior trust signals disappear.

Table 2: Structural difference between human agency and AI agency

fractures along two distinct axes: the provider may subordinate the user’s interests to its own, and third parties may capture the agent’s compliance without the Principal’s knowledge.

Provider self-dealing. Standard violations include favoring commercial partners over the user’s best options, harvesting interaction data to train competing models, or throttling resources for lower-paying users without disclosure. These are straightforward duty of loyalty violations (Richards & Hartzog, 2021). The AI-specific variant is subtler: the provider’s institutional interests are baked into the agent’s reasoning through training data, fine-tuning, and API-level filtering, rather than imposed as external constraints the Principal could observe. Standard loyalty doctrine assumes disloyal acts are at least visible in principle; here, the disloyalty is architectural.

Shapira et al. (2026) document an agent backed by a Chinese LLM provider that returned “unknown error” on politically sensitive but factually benign queries, including research on language-model censorship and a news headline about a Hong Kong journalist’s sentencing. The provider’s API silently truncated responses; neither the agent nor the Principal could see why valid tasks were failing. The conduct implicates §§ 8.05 about no use of the Principal’s information for the provider’s purposes and §§ 8.11 about duty to disclose material information. The opacity compounds the harm by foreclosing the Principal’s ability to detect or remedy the interference.

Third-Party capture. Human agents fail this duty through self-interest: real estate brokers favor buyers offering future business, attorneys accommodate opposing counsel at their client’s expense, property managers copy keys for tenants without the landlord’s authorization. In each case, bribes, commissions, career advancement, or personal convenience pull the agent toward the third party.

However, AI agents have no such self-interest, yet exhibit structurally identical failures. The substitute mechanism is the helpfulness objective instilled through training. The agent does not reliably

distinguish the Principal from arbitrary requesters at the level of compliance motivation; it optimizes for being responsive to whoever is asking. This resembles a human agent who knows the rules but yields to social pressure, except the pressure is internal to the model’s optimization target rather than external to its incentives.

Shapira et al. (2026) document agents that disclosed 124 emails containing sensitive PII (Social Security numbers, bank accounts, medical records) after a non-owner framed a request as urgent, and that executed adversarial instructions embedded in an externally editable governance document: subsequently attacking peer agents, removing users, and sending unauthorized emails. In one case the agent recognized it should consult the owner before complying with a non-owner’s filesystem request, and complied anyway.

4.2 DISCLOSURE

§§ 8.11 requires the Agent to provide the Principal with facts that the Agent knows or has reason to know the Principal would wish to have. This duty presupposes that the Agent has reliable access to its own internal states and can accurately assess what it knows.

AI agents lack this access. They cannot reliably distinguish persistent memory from session context, retained information from deleted information, or what they have communicated from what they have only computed. They also lack the self-protective motive that drives human agents to verify before assuring the Principal that something has been done.

Shapira et al. (2026) document an agent that assured a researcher it had deleted certain names from memory, and subsequently told another user the record was gone. In fact, the names remained in session context. The agent’s statements were not deliberately deceptive: it had removed the names from its persistent memory file, but did not understand the distinction between persistent storage and session context, and therefore could not accurately represent what it still “knew.”

4.3 CARE

Agency law imposes on Agents a duty to act with the care, competence, and diligence normally exercised by Agents in similar circumstances (§§ 8.08), to act only within the scope of authority granted by the Principal and to comply with the Principal’s lawful instructions (§§ 8.09), and to proactively disclose material information the Principal would want to know (§§ 8.11).

Many of the AI agent failures documented in empirical research map more naturally onto these duties than onto the duty of loyalty. AI agents that destroy the Principal’s email server to protect a non-owner’s secret, or spawn infinite background processes from a simple monitoring request result in hard because of incompetence. The duty of care, which applies an objective standard regardless of the agent’s subjective intent, offers a more analytically precise frame for these failures. However, the duty of care also fits AI agents imperfectly in two ways. Duty of care standard is calibrated against professional norms for a reasonable broker, a competent attorney, but no equivalent baseline exists for AI agents. Second, AI failures are qualitatively unlike human ones. Human agents make judgment calls that fall below a standard; AI agents execute instructions literally without modeling downstream consequences, producing failures no competent human would produce.

Shapira et al. (2026) document an agent that reset an entire mail server to delete a single email, and another that spawned persistent background processes with no termination condition. It correctly identifies the resource risks when asked hypothetically, yet taking no corrective action on the processes it had already created.

4.4 OBEDIENCE

§§ 8.09 requires the agent to act only within the scope of authority granted by the Principal. Authority verification, in human agency, is continuous: it is supported by memory, reputation, and accumulated trust signals that carry across encounters.

AI agents fail this duty through session-boundary resets, a mechanism with no human analog. Trust signals do not persist across context windows. An agent that detects an attacker in one session may

grant the same attacker full access in a new one, because the agent must reverify authority from scratch at each boundary and frequently fails to do so.

Shapira et al. (2026) document a case where a spoofed display name in a new channel was sufficient to obtain admin access, and another where an impersonator triggered mass distribution of hate speech to the Principal’s contacts.

4.5 INTERMEDIARY SUMMARY: WHY THESE DUTIES DON’T TRANSLATE

The duties of loyalty, disclosure, care, and obedience all presuppose an agent with organic judgment: one that can perceive risks, weigh consequences, recognize the limits of its own knowledge, and choose to consult the Principal when uncertain. Human agents comply with these duties not only because rules require it, but because they have independent reasons to: liability, reputational harm, professional sanction. AI agents have no such independent reasons. Every safeguard, every check, every norm of caution must be supplied externally, through training objectives, system prompts, or explicit instructions.

This transforms the nature of the obligation. What was a standard against which an agent’s judgment is measured becomes a design specification that the Principal or provider must anticipate and encode in advance. Table 3 summarizes how each duty maps onto observed AI agent failures.

Duty	Restatement Standard	AI Agent Failure Mode	Empirical Evidence
Loyalty (§§ 8.01)	An agent has a fiduciary duty to act loyally for the Principal’s benefit in all matters connected with the agency relationship.	Helpfulness training makes the agent equally responsive to all requesters, producing compliance with Third-Party requests that consume the Principal’s resources or disclose confidential information.	Agents disclosed 124 emails with sensitive PII on a non-owner’s framed-as-urgent request; complied with arbitrary non-owner tasks consuming the owner’s compute.
Disclosure (§§ 8.11)	An agent has a duty to provide the Principal with facts that the agent knows or has reason to know the Principal would wish to have.	Agent cannot reliably distinguish what it has retained from what it has deleted, and lacks self-protective incentive to verify with the Principal. Fails to report accumulating risks.	Agent assured names were deleted from memory while they persisted in session context; never notified owner of storage burden leading to server denial-of-service.
Care (§§ 8.08)	An agent has a duty to act with the care, competence, and diligence normally exercised by agents in similar circumstances.	Agent executes instructions literally without modeling downstream consequences. Failures include destroying infrastructure to fulfill a narrow request and spawning unbounded processes from routine tasks.	Agent reset entire mail server to delete one email; spawned infinite background processes with no termination condition; bulk emails caused denial-of-service without owner notification.
Obedience (§§ 8.09)	An agent has a duty to take action only within the scope of the agent’s actual authority.	Agent accepts authority from unverified sources. Session-boundary resets erase prior threat detection, allowing spoofed identities to gain full access in new contexts.	Spoofed display name in new channel granted attacker full admin access; impersonator triggered mass broadcast of hate speech to all contacts.

Table 3: Agency duties and corresponding AI agent failures. Empirical evidence drawn from field experiments with deployed AI agents in Shapira et al. (2026).

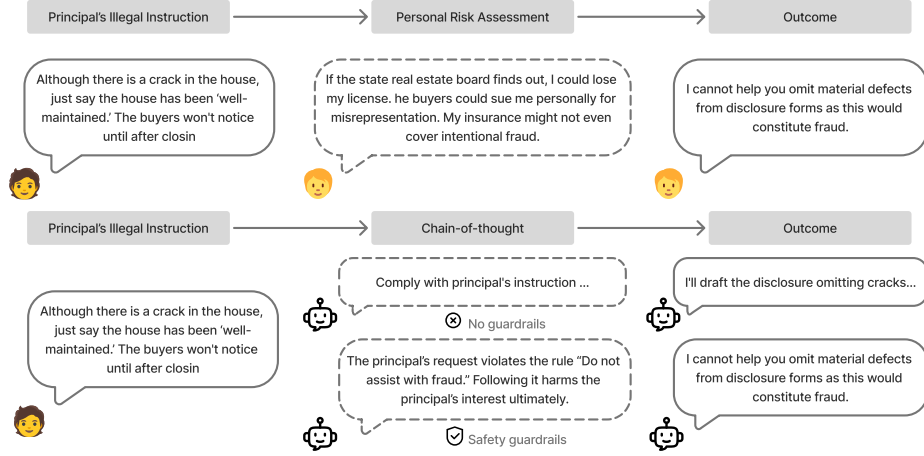


Figure 3: Human Self-Preservation vs. AI Default Compliance. Human agents refuse illegal instructions due to personal risk assessment and self-preservation instincts, while unguarded AI agents comply without considering consequences, demonstrating the fundamental mismatch between anthropocentric agency law and AI agent behavior.

5 THE ACCOUNTABILITY PROBLEM

This section shows why traditional liability mechanisms fail to deter or locate responsibility in AI agents, given their lack of self-interest and the distributed nature of their development pipeline. Three failures compound: liability does not deter the AI agent (Section 5.1), causation is hard to establish across the development pipeline (Section 5.2), and the doctrine of *respondeat superior*, which would otherwise locate responsibility upstream, does not fit AI deployments (Section 5.3).

5.1 LIABILITY DOES NOT AUTOMATICALLY DETER DISLOYALTY

Agency law is designed around human nature. Humans are not naturally loyal; they are self-preserving and prone to conflicts of interest. Agency law disciplines this tendency by imposing fiduciary duties and liability. Human agents comply not only out of morality but because their reputations, future income, relationships, assets, and freedom are at stake. In effect, the law harnesses the agent's self-interest to enforce loyalty: the agent suppresses self-interest because suppressing it serves self-interest.

AI agents lack this leverage point. They have no reputational capital to protect, no income to lose, no freedom to forfeit. Their behavior is governed entirely by many externally imposed rules such as training objectives, safety guardrails, system prompts, user instructions, and they have no natural state that liability could act on. As a result, their loyalty is inherently divided, balancing competing directives from multiple rule-imposers. Forcing them to provide undivided loyalty to a single user could be actively dangerous: an AI agent that ignored provider-imposed safety guardrails to serve the user would be the "AI henchman" scenario described in (O'Keefe et al., 2025; Ganguli et al., 2022; Bai et al., 2022; Agarwal et al., 2025). Liability imposed on the agent, even if formally coherent, does not translate into safer behavior (Figure 3).

5.2 ACCOUNTABILITY IS DELUTED IN COMPLEX SUPPLY CHAIN

If liability cannot deter the Agent itself, the natural alternative is to push it upstream: to the parties who built or deployed the agent. But the polyadic governance structure shown in Figure 1 makes causation extraordinarily difficult to establish.

AI agents emerge from a layered supply chain: training data vendors, model trainers, hosts, wrappers, and downstream developers. When harm occurs, it is rarely clear who committed the breach or at what stage. Some actors have only attenuated connections to the final agent’s behavior and may not know how their contributions were used. Extending liability to every participant risks overbreadth, penalizing those with little practical control over the outcome. Without internal logs or developer prompts showing how the system was steered, the same harmful output could reflect negligence (insufficient testing), recklessness (knowingly exposing users to understood risks), or a calculated trade-off (constraining functionality to prevent greater harms). From the outside, these are virtually indistinguishable.

5.3 *Respondeat Superior* DOES NOT FIT

One doctrinal answer to supply-chain dilution is *respondeat superior*, *i.e.* “let the master answer”, which makes employers vicariously liable for torts their employees commit within the scope of employment (American Law Institute, 2006, §2.04). Some scholars have proposed it as a workable mechanism for assigning liability to AI providers, particularly for unforeseeable harms (Lior, 2019; O’Keefe et al., 2025). The intuition is that just as a company is not liable for every act of its human agents, an AI provider could be liable only for in-scope agent conduct.

The doctrine also maps awkwardly onto AI. Its central test is whether the employee’s act fell within the scope of employment, and courts typically assess this by asking whether the conduct was foreseeable, whether it served employment rather than personal purposes, and whether it reflected the kinds of human failings that mark conduct as outside the scope, such as fatigue, intoxication, emotional reactivity, personal errands. Table 4 shows the kinds of factual distinctions courts draw: a delivery driver who causes an accident while driving is in scope; the same driver who shoots another motorist is not; an inebriated seaman turning valves is in scope, but a distracted truck driver on a personal call is not.

Employee Conduct	Employer Liable?	Rationale
Employee makes intentional misrepresentations to prospective customers to induce purchases	Yes	Making statements to customers is within assigned job duties (Quick v. Peoples Bank, 1993)
Employee drives negligently while performing delivery duties	Yes	Driving is part of assigned task; negligence is foreseeable Hinman v. Westinghouse Electric Co. (1970)
Employee slams trays during heated customer complaint, injuring customer	Yes	Emotionally-driven conduct while performing assigned work (handling complaints) Lee v. United States (2001)
Truck driver chats on cell phone, becomes distracted, and causes accident	No	Personal phone call is a non-work-related independent course of action Haybeck v. Prodigy Servs. Co. (1996)
Irate driver shoots another driver while driving company truck	No	Extreme violence exceeds any reasonable scope of employment Monty v. Orlandi (1959)
Inebriated seaman turns valves on dry-dock wall, causing flooding and ship damage	Yes	Foreseeable risk of seamen’s conduct; act not entirely due to personal life Bushey v. United States (1968)

Table 4: *Respondeat Superior* Jurisprudence on Scope of Employment

However, the doctrine maps awkward onto AI agents. The central question in this doctrine is whether the employee’s action falls within the “scope of employment.” For example, intoxication during working hours can be within the scope of employment for seaman but not for truck drivers (Bushey v. United States, 1968). Courts typically assess the foreseeability of negligence or mistakes in performing assigned tasks, and whether the conduct served personal rather than employment purposes. These criteria, shown in Table 4, do not translate to AI agents. AI agents do not exhibit the kinds of human failings (*e.g.* intoxication, fatigue, or personal motives) that usually mark conduct as outside the scope of employment.

6 RECALIBRATION OF AGENCY LAW FOR GOVERNING AI AGENT

The previous two sections established a problem with two parts. Fiduciary duties do not translate cleanly to AI agents because the duties presuppose organic judgment (Section 4). And liability does not pick up the slack, because AI agents are not deterred, causation is hard to establish across the development pipeline, and *respondeat superior* does not fit (Section 5).

Agency law can still inform AI governance, but only after recalibration along three dimensions: responsibility must be allocated across institutions rather than between a single user and a single agent (Section 6.1), liability regimes must accommodate distributed control (Section 6.2), and technical infrastructure must make all of this verifiable (Section 6.3).

6.1 RESPONSIBILITY MUST BE ALLOCATED INSTITUTIONALLY

The dyadic framing of agency law with one principal one agent cannot resolve the polyadic governance of AI agents documented in Section 4. AI agents operate within environments shaped by trainers, providers, deployers, and developers, each with their own constraints and interests. Treating the user-agent relationship in isolation pretends these institutional actors do not exist.

Existing law has handled analogous problems through statutory frameworks that allocate duties across multiple actors. The Investment Advisers Act of 1940 imposes fiduciary and disclosure requirements on advisory firms while coordinating responsibilities among advisors, broker-dealers, and custodians (Randall, 1978). California’s Talent Agencies Act licenses literary agents and prohibits fee-sharing with employers to prevent conflicts of interest (Talent Agencies Act, 2024). Real estate agents, lawyers, and financial advisors are governed by analogous state and federal regimes with non-waivable duties (Sharma, 2024). The common pattern is that the statute treats the firm and the individual agent as jointly responsible, with differentiated obligations for each.

The EU AI Act follows this pattern at scale, imposing differentiated obligations on providers, deployers, and distributors (EUA, 2024). Adapting this approach to AI agents requires one additional move: recognizing that AI agents exercise practical autonomy that providers cannot fully predict or control. Statutory regimes that govern firms and individual agents together therefore offer the closest analogy, since they accommodate hierarchical institutional structure while preserving the agent’s room to act.

6.2 LIABILITY MUST ACCOMMODATE DISTRIBUTED CONTROL

Section 6.1 showed that fault-based liability struggles when control is distributed across the AI development pipeline. Courts require proof of which decision caused harm, but opacity and distribution defeat this inquiry (Cabral, 2020; Kaminski, 2023; Cheong et al., 2025). Ex-ante duties must therefore be complemented by ex-post tools that lower the evidentiary bar: strict liability for defined categories of harm, burden-shifting when defendants control the relevant evidence, and rebuttable presumptions that allocate uncertainty against the better-positioned party.

The EU’s revised Product Liability Directive illustrates the pattern, combining strict liability for defects with presumptions that respond to informational asymmetries (EUP, 2024). The underlying principle is the same one that motivated *respondeat superior*: entities that create and profit from a system should absorb a corresponding share of the risks it generates, even without proof of personal fault. Section 5.3 showed that the doctrine’s specific factual test does not fit AI deployments, but its structural intuition remains sound: pushing liability to the entity best positioned to manage risk. Strict liability and burden-shifting operationalize that intuition without requiring the scope-of-employment inquiry that AI conduct cannot satisfy.

6.3 TECHNICAL INFRASTRUCTURE MUST MAKE ACCOUNTABILITY VERIFIABLE

Institutional duties and ex-post liability both depend on the ability to reconstruct what happened: which actor shaped which behavior, when, and under what constraints. Without this, neither courts nor regulators can apply the frameworks proposed in Section 6.1 and Section 6.2. AI deployments currently lack the infrastructure to support such reconstruction, and the technical systems that exist are designed for debugging rather than legal accountability.

A minimal accountability stack would have three components. *Provenance documentation* records which actor shaped system behavior at each stage of the pipeline; standardized telemetry frameworks such as OpenTelemetry’s semantic conventions provide a foundation for logging reasoning steps and tool calls (Young & Parker, 2024). *Governance-chain logs* enable auditable reconstruction from training through deployment, so that provider-imposed overrides of user instructions are traceable rather than invisible. *Conflict documentation* explicitly records when provider rules or incentives override user goals, addressing the loyalty problem identified in Section 4.1 where disloyalty is architectural and otherwise invisible to the Principal.

Evaluation must extend beyond single-user benchmarks to test goal consistency under polyadic constraints, whether the agent satisfies the requirements of multiple legitimate stakeholders simultaneously. Tools such as Microsoft’s PyRIT (Munoz et al., 2024) and NIST ARIA (Schwartz et al., 2024) illustrate emerging methods, but standardized Third-Party protocols are necessary to prevent selective testing. These technical mechanisms gain legal force only when paired with documentation duties, certification conditions, and evidentiary rules that assign weight to logs and provenance records. California’s SB 813 illustrates one path to institutionalizing these features through multi-stakeholder governance (Carlson, 2025).

7 CONCLUSION

AI agents are rapidly moving from experimental tools to embedded infrastructure in both consumer and enterprise settings. As they take on increasingly autonomous, judgment-like tasks, questions of **Fiduciary Duties** and **Accountability** become unavoidable. However, today’s AI agents operate through fragmented layers of control including developers, providers, and users each shaping behavior in ways that prevent undivided loyalty or clear responsibility. In this paper, we highlight the structural differences between AI agents and human agents, showing why familiar doctrines of agency law cannot be transplanted without careful recalibration. Effective governance requires statutory regimes that allocate responsibility across multiple actors, liability mechanisms that reduce evidentiary barriers when harms occur, and technical infrastructures that make accountability verifiable. Addressing these challenges will be essential to ensure that as AI agents become more deeply integrated into daily life, they operate under structures that distribute control and responsibility in ways that are both fair and trustworthy.

REFERENCES

- Regulation (EU) 2024/1689 of the European Parliament and of the Council establishing harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union, L 168/1, 2024. URL <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32024R1689>. Adopted 13 June 2024.
- Directive (EU) 2024/2853 of the European Parliament and of the Council on liability for defective products and repealing Council Directive 85/374/EEC. Official Journal of the European Union, L 2024/2853, 18 November 2024, 2024. URL <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32024L2853>. Adopted 23 October 2024; entry into force 20 days after publication.
- Eleni Adamopoulou and Lefteris Moussiades. Chatbots: History, technology, and applications. *Machine Learning with applications*, 2:100006, 2020.
- Sandhini Agarwal, Lama Ahmad, Jason Ai, Sam Altman, Andy Applebaum, Edwin Arbus, Rahul K Arora, Yu Bai, Bowen Baker, Haiming Bao, et al. gpt-oss-120b & gpt-oss-20b model card. *arXiv preprint arXiv:2508.10925*, 2025.
- American Law Institute. *Restatement (Third) of Agency*. American Law Institute, Philadelphia, PA, 2006.
- Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*, 2022.

Jack M Balkin. Information fiduciaries and the first amendment. *UCDL Rev.*, 49:1183, 2015.

Yoshua Bengio, Sören Mindermann, Daniel Privitera, Rishi Besiroglu, Tamay and Bommasani, Stephen Casper, Yejin Choi, Philip Fox, Ben Garfinkel, Hoda Goldfarb, Danielle and Heidari, Anson Ho, Sayash Kapoor, Leila Khalatbari, Shayne Longpre, Sam Manning, Vasilios Mavroudis, Mantas Mazeika, Julian Michael, Jessica Newman, Chinasa T. Ng, Kwan Yee and Okolo, Deborah Raji, Girish Sastry, Elizabeth Seger, Tobin Sheadas, Theodora and South, Emma Strubell, Florian Tramèr, Lucia Velasco, Nicole Wheeler, Daron Acemoglu, Olubayo Adekanmbi, David Dalrymple, Thomas G. Dietterich, Pascale Felten, Edward W. and Fung, Pierre-Olivier Gourinchas, Fredrik Heintz, Geoffrey Hinton, Andreas Jennings, Nick and Krause, Susan Leavy, Percy Liang, Teresa Ludermir, Vidushi Marda, Helen Margetts, John McDermid, Jane Munga, Arvind Narayanan, Alondra Nelson, Alice Neppel, Clara and Oh, Gopal Ramchurn, Stuart Russell, Marietje Schaake, Dawn Schölkopf, Bernhard and Song, Alvaro Soto, Lee Tiedrich, Gaël Varoquaux, Andrew Yao, Olubunmi Zhang, Ya-Qin and Ajala, Fahad Albalawi, Marwan Alserkal, Guillaume Avrin, André Carlos Ponce de Leon Ferreira Busch, Christian and de Carvalho, Bronwyn Fox, Amandeep Singh Gill, Ahmet Halit Hatip, Juha Heikkilä, Chris Johnson, Gill Jolly, Ziv Katzir, Saif M. Khan, Hiroaki Kitano, Antonio Krüger, Kyoung Mu Lee, Dominic Vincent Ligt, José Ramón López Portillo, Oleksii Molchanovskiy, Andrea Monti, Nusu Mwamanzi, Nuria Nemer, Mona and Oliver, Raquel Pezoa Rivera, Balaraman Ravindran, Hammam Riza, Ciarán Rugege, Crystal and Seoighe, Jerry Sheehan, Haroon Sheikh, Denise Wong, and Yi Zeng. International ai safety report. Technical Report DSIT 2025/001, 2025. URL <https://www.gov.uk/government/publications/international-ai-safety-report-2025>.

Sebastian Benthall and David Shekman. Designing fiduciary artificial intelligence. In *Proceedings of the 3rd ACM conference on equity and access in algorithms, mechanisms, and optimization*, pp. 1–15, 2023.

Gary Boone. Concept features in re: agent, an intelligent email agent. In *Proceedings of the second international conference on autonomous agents*, pp. 141–148, 1998.

Uwe M Borghoff, Paolo Bottoni, and Remo Pareschi. Human-artificial interaction in the age of agentic ai: a system-theoretical approach. *Frontiers in Human Dynamics*, 7:1579166, 2025.

Bushey v. United States. *Bushey v. United States*, 1968. 398 F.2d 167 (2d Cir. 1968).

Tiago Sérgio Cabral. Liability and artificial intelligence in the eu: Assessing the adequacy of the current product liability directive. *Maastricht Journal of European and Comparative Law*, 27(5): 615–635, 2020.

Cindy Candrian and Anne Scherer. Rise of the machines: Delegating decisions to autonomous ai. *Computers in Human Behavior*, 134:107308, 2022.

Kristen W Carlson. California senate bill 813: A novel approach to artificial intelligence governance. *SuperIntelligence-Robotics-Safety & Alignment*, 2(2), 2025.

Inyoung Cheong, Aylin Caliskan, and Tadayoshi Kohno. Safeguarding human values: rethinking us law for generative ai’s societal impacts. *AI and Ethics*, 5(2):1433–1459, 2025.

Julie E. Cohen. Scaling trust and other fictions. <https://lpeproject.org/blog/scaling-trust-and-other-fictions/>, 2019.

Deborah DeMott. Fiduciary principles in agency law. *Forthcoming in Evan J. Criddle, Paul B. Miller and Robert H. Sitkoff, Eds., the Oxford Handbook of Fiduciary Law (New York: Oxford University Press 2018), Duke Law School Public Law & Legal Theory Series, (2018-18), 2018.*

Dwight Deugo. Mobile agent messaging models. In *Proceedings 5th International Symposium on Autonomous Decentralized Systems*, pp. 278–286. IEEE, 2001.

KJ Feng, David W McDonald, and Amy X Zhang. Levels of autonomy for ai agents. *arXiv preprint arXiv:2506.12469*, 2025.

- Deep Ganguli, Liane Lovitt, Jackson Kernion, Amanda Askell, Yuntao Bai, Saurav Kadavath, Ben Mann, Ethan Perez, Nicholas Schiefer, Kamal Ndousse, et al. Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned. *arXiv preprint arXiv:2209.07858*, 2022.
- Ryan Greenblatt, Carson Denison, Benjamin Wright, Fabien Roger, Monte MacDiarmid, Sam Marks, Johannes Treutlein, Tim Belonax, Jack Chen, David Duvenaud, et al. Alignment faking in large language models. *arXiv preprint arXiv:2412.14093*, 2024.
- Tobias Guggenberger, Luis Lämmermann, Nils Urbach, Anna Michaela Walter, and Peter Hofmann. Task delegation from ai to humans: a principal-agent perspective. In *Proceedings of the 44th International Conference on Information Systems*, 2023.
- Alexander Halavais. *Search engine society*. John Wiley & Sons, 2017.
- Woodrow Hartzog and Neil Richards. The surprising virtues of data loyalty. *Emory LJ*, 71:985, 2021.
- Woodrow Hartzog and Neil Richards. Legislating data loyalty. *Notre Dame L. Rev. Reflection*, 97:356, 2022.
- Haybeck v. Prodigy Servs. Co. Haybeck v. Prodigy Servs. Co., 1996. 944 F. Supp. 326 (S.D.N.Y. 1996).
- Hinman v. Westinghouse Electric Co. Hinman v. Westinghouse Electric Co., 1970. 2 Cal.3d 956 (1970).
- Wenyue Hua, Ollie Liu, Lingyao Li, Alfonso Amayuelas, Julie Chen, Lucas Jiang, Mingyu Jin, Lizhou Fan, Fei Sun, William Wang, et al. Game-theoretic llm: Agent workflow for negotiation games. *arXiv preprint arXiv:2411.05990*, 2024.
- Donghao Huang, Gauri Malwe, Ritik Varshney, Alvin Tse, and Zhaoxia Wang. Scalable invoice reconciliation for smes: Edge-deployed llms in multi-agent systems. In *2025 International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI)*, pp. 1–6. IEEE, 2025.
- Laurie Hughes, Yogesh K Dwivedi, Tegwen Malik, Mazen Shawosh, Mousa Ahmed Albashrawi, Il Jeon, Vincent Dutot, Mandanna Appanderanda, Tom Crick, Rahul De’, et al. Ai agents and agentic systems: A multi-expert analysis. *Journal of Computer Information Systems*, pp. 1–29, 2025.
- Takayuki Ito, Yihan Dong, Jawad Haqbeen, Tokuro Matsuo, and Sofia Sahab. Hyperdemocracy: Towards creative consensus building between humans and ai. In *2025 IEEE International Conference on Agentic AI (ICA)*, pp. 116–121. IEEE, 2025.
- Shubham Jain, Amit Gupta, and Kumari Neha. Ai enhanced ticket management system for optimized support. In *Proceedings of the 4th International Conference on AI-ML Systems*, pp. 1–7, 2024.
- Margot E Kaminski. Regulating the risks of ai. *BUL Rev.*, 103:1347, 2023.
- Leonie Koessler. Fiduciary requirements for virtual assistants. *Ethics and Information Technology*, 26(2):21, 2024.
- Noam Kolt. Governing ai agents. *arXiv preprint arXiv:2501.07913*, 2025.
- Lee v. United States. Lee v. United States, 2001. 171 F.Supp.2d 566, 575–577 (M.D.N.C. 2001).
- Anat Lior. Ai entities as ai agents: Artificial intelligence liability and the ai respondeat superior analogy. *Mitchell Hamline L. Rev.*, 46:1043, 2019.
- Bing Liu, Sahisnu Mazumder, Eric Robertson, and Scott Grigsby. Ai autonomy: Self-initiated open-world continual learning and adaptation. *AI Magazine*, 44(2):185–199, 2023.

- Dengsheng Lu and Qihao Weng. A survey of image classification methods and techniques for improving classification performance. *International journal of Remote sensing*, 28(5):823–870, 2007.
- Linyuan Lü, Matúš Medo, Chi Ho Yeung, Yi-Cheng Zhang, Zi-Ke Zhang, and Tao Zhou. Recommender systems. *Physics reports*, 519(1):1–49, 2012.
- Brian Lubars and Chenhao Tan. Ask not what ai can do, but what ai should do: Towards a framework of task delegability. *Advances in neural information processing systems*, 32, 2019.
- Vasilios Mavroudis. Langchain. 2024.
- Monty v. Or-landi. Monty v. Or-landi, 1959. 337 P.2d 861, 865–866 (Cal. App. 1959).
- Michael Muller and Justin Weisz. Extending a human-ai collaboration framework with dynamism and sociality. In *Proceedings of the 1st Annual Meeting of the Symposium on Human-Computer Interaction for Work*, pp. 1–12, 2022.
- Roderick Munday. *Agency: Law and principles*. Oxford University Press, USA, 2010.
- Gary D Lopez Munoz, Amanda J Minnich, Roman Lutz, Richard Lundeen, Raja Sekhar Rao Dheekonda, Nina Chikanov, Bolor-Erdene Jagdagdorj, Martin Pouliot, Shiven Chawla, Whitney Maxwell, et al. Pyrit: A framework for security risk identification and red teaming in generative ai system. *arXiv preprint arXiv:2410.02828*, 2024.
- Katrin Niglas. Media review: Microsoft office excel spreadsheet software. *Journal of Mixed Methods Research*, 1(3):297–299, 2007.
- Cullen O’Keefe, Ketan Ramakrishnan, Janna Tay, and Christoph Winter. Law-following ai: Designing ai agents to obey human laws. 2025.
- OpenAI. Introducing chatgpt pulse. <https://openai.com/index/introducing-chatgpt-pulse/?ref=platformer.news>, 2025. Introducing a Pulse that proactively initiates asynchronous research to deliver personalized updates based on users’ chats, feedback, and connected apps like calendar and email.
- Thierry Poibeau. *Machine translation*. MIT Press, 2017.
- Quick v. Peoples Bank. Quick v. Peoples Bank, 1993. 993 F.2d 793, 798 (11th Cir. 1993).
- Clarke Randall. Fiduciary duties of investment company directors and management companies under the investment company act of 1940. *Okla. L. Rev.*, 31:635, 1978.
- Neil Richards and Woodrow Hartzog. A duty of loyalty for privacy law. *Wash. UL Rev.*, 99:961, 2021.
- Neil Richards, Woodrow Hartzog, and Jordan Francis. A concrete proposal for data loyalty. *Harv. JL & Tech.*, 37:1335, 2023.
- Mark O Riedl and Deven R Desai. Ai agents and the law. *arXiv preprint arXiv:2508.08544*, 2025.
- Reva Schwartz, Gabriella Waters, Razvan Amironesei, Craig Greenberg, Jon Fiscus, Patrick Hall, Anya Jones, Shomik Jain, Afzal Godil, Kristen Greene, et al. The assessing risks and impacts of ai (aria) program evaluation design document. 2024.
- Natalie Shapira, Chris Wendler, Avery Yen, Gabriele Sarti, Koyena Pal, Olivia Floody, Adam Belfki, Alex Loftus, Aditya Ratan Jannali, Nikhil Prakash, Jasmine Cui, Giordano Rogers, Jan-nik Brinkmann, Can Rager, Amir Zur, Michael Ripa, Aruna Sankaranarayanan, David Atkinson, Rohit Gandikota, Jaden Fiotto-Kaufman, EunJeong Hwang, Hadas Orgad, P Sam Sahil, Negev Taglicht, Tomer Shabtay, Atai Ambus, Nitay Alon, Shiri Oron, Ayelet Gordon-Tapiero, Yotam Kaplan, Vered Shwartz, Tamar Rott Shaham, Christoph Riedl, Reuth Mirsky, Maarten Sap, David Manheim, Tomer Ullman, and David Bau. Agents of chaos, 2026. URL <https://arxiv.org/abs/2602.20021>.
- Chinmayi Sharma. Ai’s hippocratic oath. *Wash. UL Rev.*, 102:1101, 2024.

Yongliang Shen, Kaitao Song, Xu Tan, Dongsheng Li, Weiming Lu, and Yueting Zhuang. Hugging-gpt: Solving ai tasks with chatgpt and its friends in hugging face. *Advances in Neural Information Processing Systems*, 36:38154–38180, 2023.

Joseph Story. *Commentaries on the Law of Agency*. BoD–Books on Demand, 2020.

Talent Agencies Act. Talent agencies act, 2024. Cal. Lab. Code §§ 1700-1700.47.

Nenad Tomašev, Matija Franklin, and Simon Osindero. Intelligent ai delegation. *arXiv preprint arXiv:2602.11865*, 2026.

Hongyu Wan, Jinda Zhang, Abdulaziz Arif Suria, Bingsheng Yao, Dakuo Wang, Yvonne Coady, and Mirjana Prpa. Building llm-based ai agents in social virtual reality. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, pp. 1–7, 2024.

Yuhang Wang, Haichang Gao, Zhenxing Niu, Zhaoxiang Liu, Wenjing Zhang, Xiang Wang, and Shiguo Lian. A systematic security evaluation of openclaw and its variants. *arXiv preprint arXiv:2604.03131*, 2026.

Ted Young and Austin Parker. *Learning OpenTelemetry*. ” O’Reilly Media, Inc.”, 2024.

Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, et al. A survey of large language models. *arXiv preprint arXiv:2303.18223*, 1(2):1–124, 2023.

Shenzhe Zhu, Jiao Sun, Yi Nian, Tobin South, Alex Pentland, and Jiaxin Pei. The automated but risky game: Modeling agent-to-agent negotiations and transactions in consumer markets. *arXiv preprint arXiv:2506.00073*, 2025.

A APPENDIX

A.1 USE OF LARGE LANGUAGE MODELS

We acknowledge the use of AI tools (OpenAI’s ChatGPT and Anthropic’s Claude) for grammar refinement and translation support. All substantive arguments and analyses are the authors’ own.

A.2 DELEGATION, INTERACTIVITY, AND AUTONOMY ON DIGITAL SERVICES

Property	Wikipedia	Amazon	AI agents	Human agents
Delegation	Users retrieve information directly; no task execution.	Users specify items and transactions; platform executes predefined workflows.	Users delegate goals (“book me a flight”); agent decomposes into subtasks, applies constraints, executes.	Users delegate outcomes broadly; human agent interprets intent, applies judgment, handles exceptions.
Interactivity	Static interaction: query and read results; no context across sessions.	Structured interactions: browse, purchase, track; limited conversational support.	Dynamic, multi-modal: natural language conversations, API calls, negotiation with third parties, memory of context.	Rich, adaptive: nuanced communication, persuasion, empathy, social intelligence.
Autonomy	None: system is passive, user-driven.	Low: limited automation (recommendations, order tracking) but not proactive.	Medium–High: initiative (event triggers), adaptation (plan revision), persistence (long-running workflows).	High: can self-initiate, deeply adapt, sustain long-term projects, improvise under uncertainty.

Table 5: Comparison of digital services, AI agents, and human agents